Journal of Archaeological Method and Theory (2026) 33:23
https://doi.org/10.1007/s10816-025-09757-x

RESEARCH

Differential Taphonomic Behavior of Flint Types: Experimental Insights and Implications for the Middle Stone Age Assemblages of Wadi Lazalim in the Northern Sahara (Kebili, Tunisia)

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Received: 12 June 2025 / Accepted: 27 November 2025
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Abstract

Several syn and post-depositional processes are responsible for different degrees of site and assemblage formation and disturbance. Understanding the processes that lead to archaeological site formation is essential for interpreting past human activities, settlement patterns, and occupation dynamics. Among these processes, water flow is commonly recognized as a major factor influencing site and assemblage formation and integrity. Lithic taphonomy can add valuable information to the understanding of these natural processes. To better evaluate how different flint types react to both mechanical and chemical stresses, we carried out controlled and sequential experiments on knapped flint artifact replicas. These were realized on two different flint varieties, recognized in the archaeological assemblage of a Middle Stone Age open-air site located at Wadi Lazalim (Southern Tunisia). Our methodological framework is based on a sequential tumbling experiment that simulates the remobilization of artifacts by water. The resulting post-depositional surface modifications (PDSM) were observed and recorded. In addition, a sample of flint replicas underwent chemical alteration in a controlled environment, allowing us to document and measure both patina formation and changes in surface roughness. These experimental results were then compared with a sample of archaeological materials, providing the basis for a preliminary and exploratory hypothesis concerning the taphonomic processes at the site. Here, we observed that most of the alterations appear to be linked to patina formation, whose variable degree and kind are more strongly correlated with differences in raw material texture rather than to the co-occurrence of artefacts from different chronological spans.

Keywords Lithic taphonomy · Flint alteration · Experimentation · Confocal microscopy · Tunisia

Extended author information available on the last page of the article

Published online: 16 December 2025

Introduction

Since the pioneering work of Michael B. Schiffer on the identification of site formation processes (Schiffer, 1972, 1983, 1987), a large body of studies has deepened our understanding of the drivers that contribute to site formation, alteration, and preservation, with important implications for the understanding of past human settlement patterns, cultural behaviors, and activities (*e.g.*, Berger, 2011; Binford, 1979, 1980, 1981; Davies & Holdaway, 2017; Friesem *et al.*, 2014; Janovský *et al.*, 2022; Luria *et al.*, 2020; Masoj´c *et al.* 2024; Martínez *et al.*, 2019; Moncel *et al.*, 2023; Navazo & Díez, 2008; Pilø *et al.*, 2021). Among the natural processes involved in site formation, water flow is considered one of the most common in Palaeolithic/Stone Age archaeology worldwide (*e.g.*, Bertran *et al.*, 2019; Chu, 2016; Chu & Hosfield, 2020; Delagnes *et al.*, 2023; Domínguez-Rodrigo *et al.*, 2014; Ferring, 2001; García-Moreno *et al.*, 2016; Hosfield, 2011; Hosfield & Chambers, 2005, 2016; Hosfield *et al.*, 2000; Hovers, 2003; Petraglia & Potts, 1994; Schick, 1986; Shackley, 1974; Shea, 1999; Sitzia *et al.*, 2012). During water transport, lithics can experience mechanical alterations due to particle—and artifacts themselves—remobilization, resulting in macro- and mesoscopic wear (edge/ridge rounding, fractures, flaking, abrasion) as well as in microscopic ones (striations, polish) (Asryan *et al.*, 2014; Bustos-Pérez *et al.*, 2019; Bustos-Pérez & Ollé, 2024a; Chu & Hosfield, 2020; Hosfield & Chambers, 2005, 2016; Levi-Sala, 1986; Schoville *et al.*, 2016). Such traces are highly recognizable since they have specific morphologies and random orientations and distributions over the artifact's surfaces (Asryan *et al.*, 2014; Burroni *et al.*, 2002; Mazzucco, 2014; Tringham *et al.*, 1974; Wilkins & Schoville, 2016) and can be easily distinguished from use-wear (Lemorini *et al.*, 2014; Zupancich *et al.*, 2016). In addition, patina formation, which is a gradual chemical process involving desilication and silica recrystallization or just desilication (Caux *et al.*, 2018; Fiers, 2020; Rottländer, 1975), can significantly affect and alter the lithic surface and structure. Both mechanical and chemical processes are indeed closely tied to the specific environmental context in which the artifacts were buried (Burroni *et al.*, 2002; Thiry *et al.*, 2014). Although time-consuming, the study of lithic taphonomic behaviors is essential for determining the degree and intensity of the development of post-depositional surface modifications (henceforth PDSM) (Bustos-Pérez *et al.*, 2019; Bustos-Pérez & Ollé, 2024a; Caux *et al.*, 2018; Chambers, 2004, 2016; Chu *et al.*, 2015; Glauberman & Thorson, 2012; Howard, 2002; Shea, 1999; Thiry *et al.*, 2014). Through a macro- and microscopic analysis of lithics and applying both qualitative and quantitative recording of PDSM to assess the degree of alteration, research has provided a solid foundation for better interpreting site formation processes and assemblage integrity (Asryan *et al.*, 2014; Berruti & Arzarello, 2020; Bustos-Pérez & Ollé, 2024a; Chu *et al.*, 2015; Galland *et al.*, 2019, 2025; Levi-Sala, 1986; Shackley, 1978; Petraglia & Potts, 1994; Pevny, 2012). The setting of experimental reference collections, with experiments carried out both in the laboratory and in the field, has greatly deepened our understanding of how flint (Burroni *et al.*, 2002; Bustos-Pérez & Ollé, 2024a; Eren *et al.*, 2011; Levi-Sala, 1986; Mazzucco *et al.*, 2013; Michel *et al.*, 2019), quartz and quartzite (Bustos-Pérez *et al.*,

2019; Galland *et al.*, 2025; Ollé *et al.*, 2016; Venditti *et al.*, 2016; Weitzel *et al.*, 2014), and volcanic rocks (Asryan *et al.*, 2014; Brito-Abrante *et al.*, 2025; Clemente & Gibaja, 2009; Kononenko, 2011) alter, providing a valuable tool for the study of lithic taphonomy.

A recently excavated late Middle Pleistocene open-air site at Wadi Lazalim, in southern Tunisia, has yielded abundant lithics, confirming occupation of the region during the Early Middle Stone Age (Cancellieri *et al.*, 2022). Based on the preliminary recognition of some PDSM and the stratigraphic nature of the deposit, the lithic material recovered at the site, exclusively made out of two flint varieties, is thought to have suffered from some degree of remobilization and chemical surface alteration. We thus performed laboratory experiments on artifact replicas made of the two varieties of flint, aimed at evaluating the following: (i) the mechanical effects of hydraulic transport on the development of PDSM and (ii) the degree of surface alteration prompted by patina formation. A mesoscopic and microscopic record of the PDSM was carried out, and changes in surface roughness were measured using a confocal microscope. The results were compared with a sample of archaeological artifacts recovered from site 16/15 at Wadi Lazalim. The experiments highlight that the two different flint types react differently to taphonomic processes (both mechanical and chemical), constituting an exploratory framework for a preliminary interpretation of post-depositional and formation processes at the site.

Materials and methods

The site

The area of Wadi Lazalim is located ~ 40 km east of the city of Douz, in the Kebili region in southern Tunisia (Fig. 1A). It is a predominantly flat area with small hills (from ~ 50 up to 100 m above sea level) and numerous wadis. The Wadi Lazalim watershed (51.4 km^2) drains part of the Jebel Daouaia and the hills on its west side. The area has been investigated since 2015 by the Institut National du Patrimoine of Tunis and the Archaeological Mission in the Sahara of Sapienza University of Rome, and has led to the identification of several Pleistocene sites (di Lernia *et al.*, 2017). Two open-air stratified sites (15/1, 16/29) were excavated along the banks of the Wadi Lazalim and two (sites 16/15 and 19/1) along the right banks of one of its minor tributaries (Fig. 1C). Geomorphological and sedimentary analyses suggested the secondary nature of the deposits, which formed in semi-arid environmental conditions by alluvial-colluvial deposition processes and surface runoff (Cancellieri *et al.*, 2022). The sedimentation was thus driven by slope processes, alternated with alluvial and erosional ones, with the occasional activation/deactivation of the former throughout the alternation of interglacial/glacial–humid/arid phases during the Middle and Upper Pleistocene, as also recorded elsewhere in the Sahara and its surroundings (Causse *et al.*, 2003; Zerboni *et al.*, 2011). Finally, pedogenetic events characterize the youngest sedimentary units, where progressive carbonate depletion and gypsum enrichment also witness a trend towards arid conditions.

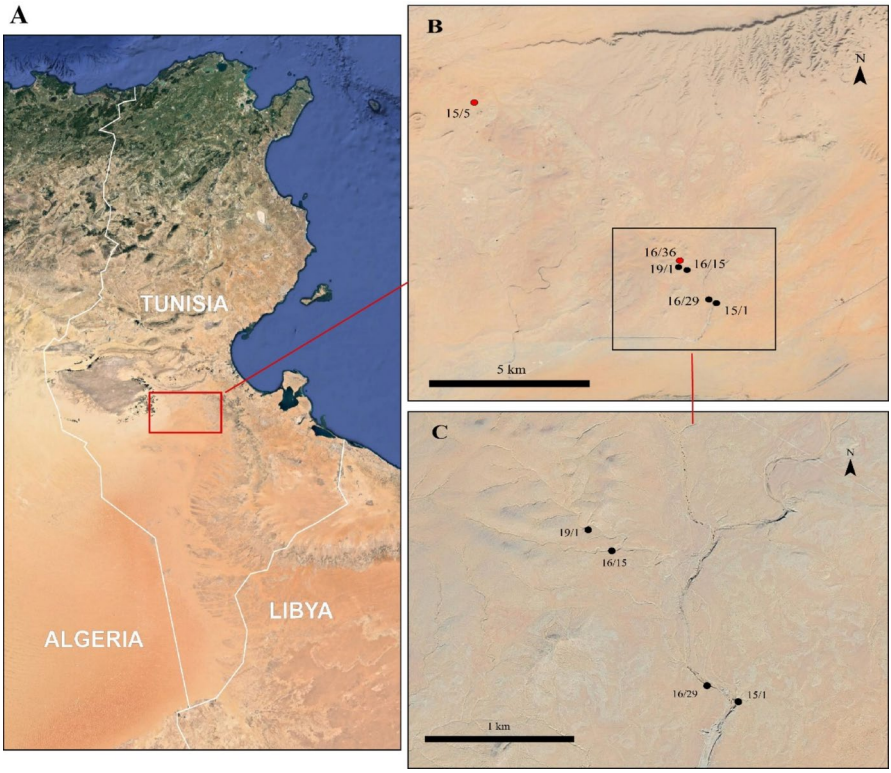


Fig. 1 Maps showing (**A**) the localization of the Wadi Lazalim area in southern Tunisia; (**B**) sites excavated (black dots) and flint outcrops (red dots) mentioned in the text; (**C**) detail of the localization of the excavated sites along the banks of the Wadi Lazalim and its tributary. Maps have been generated with Google Maps (**A**) and Google Earth Pro (**B** and **C**)

At site 16/15 (here used as the reference site for the experiments, see below), the sequence, 2.1 m deep, is characterized by 11 sedimentary units. They all have a sandy-silty matrix. From top to bottom, layer A is an unconsolidated coarse deposit, layers B and C are moderately consolidated deposits with few gravels, and layers D and F are moderately consolidated deposits with rare clasts. Layers G and H are consolidated deposits with pebbles and gravels (1–5 cm), while layers I and J are well-consolidated coarse deposits with pebbles (10–30 cm). The underlying layer K is a white limestone hard bedrock (Fig. 2). From a granulometric perspective, sand is prevalent along the sequence, with slightly coarser sand from 100 cm downwards (from layer G approximately), constituting between 40 and 70% of the sediment's component, while gravels and pebbles (less than 10 cm) represent ~ 10%. The proportions between the various components are homogeneous along the sequence, with gravels and lithics within sandy-silty sediments (Cancellieri *et al.*, 2022). The presence of millimetric carbonate granules and gypsum crystals characterizes the whole sequence. However, from

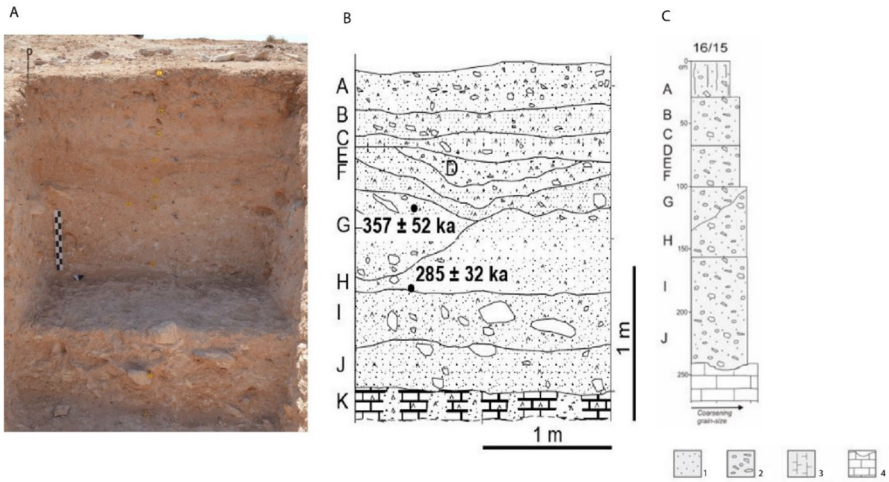


Fig. 2 South profile (**A**), stratigraphic sequence (**B**), and log (**C**) of site 16/15 with the indication of the layers described in the text (**B** and **C** modified after Cancellieri *et al.*, 2022). Key to C): (1) sandy to silty matrix; (2) coarse constituents; (3) silty to clay deposit; (4) bedrock

the lowest layers upwards, a general carbonate depletion and gypsum accumulation testify to a progressive transition towards arid conditions that fostered water evapotranspiration.

A total of 720 lithic artifacts have been recovered from site 16/15. While materials from site 19/1 are currently under study, artifacts from 16/15, together with the ones collected at site 15/1 ($N = 291$) and at site 16/29 ($N = 124$) (for a total of 1135 artifacts), testify to a Middle Stone Age (MSA) occupation of the area during the late Middle Pleistocene (MIS 8 to MIS 6) (Cancellieri *et al.*, 2022). Different reduction methods were used (mainly the Levallois, but also, and to a lesser extent, the laminar, the Kombewa, and the Discoid) to produce flakes and blades. Scrapers, end-scrapers, and denticulates are the most attested tools in the overall sequences; some bifacial tools from site 16/15, assimilated to core-axes, testify to a heavy-duty component that was part of the earliest assemblages (Cancellieri *et al.*, 2022). Artifacts were realized on two different varieties of flint, recognized macroscopically: a brownish coarse-grained one and a vitreous, fine-grained multicolor one (Fig. 3). The source of the former was identified in a natural outcrop of upper Cretaceous age, less than 1 km far from site 16/15 (outcrop 16/36), while possible sources for the finest variety have been found 10 km northward from the sites (outcrop 15/5) (Fig. 1B). A preliminary macroscopic analysis showed the presence of different degrees of mechanical and chemical alteration of the lithics, which have been grouped as follows: Group 1 is composed of coarse to medium-grained flint artifacts with a high degree of surface alteration (white patina); Group 2 is composed of medium to fine-grained flint moderately affected by chemical alteration in the form of a white or gray patina; Group 3 is composed of very fine-grained flint with moderate to no presence of patina (Cancellieri *et al.*, 2022).

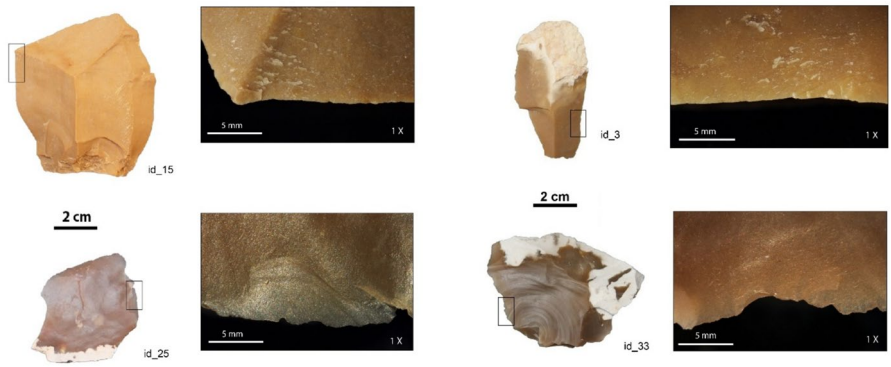


Fig. 3 Selection of artifact replicas made from two varieties of flint: a coarse-grained one (id_15 and id_3) and a fine-grained one (id_25 and id_33) (the complete sample of artifact replicas analyzed and used in the experiment, along with their morphometric and technological characteristics, is presented in Table 1). The fresh state of edges and surfaces was photographically documented before the experiments

Raw material and experimental artifacts

First, geological samples of the two varieties of flint recorded in the archaeological assemblage of site 16/15 were collected from the two main outcrops identified in the Wadi Lazalim area (Fig. 1). Four blocks of brownish flint were collected in an area spanning 100 m from site 16/15, while two blocks of darkish flint were collected from the outcrop 15/5. Future petrographic analyses on both the geological and archaeological samples will allow for a better characterization of the two flint varieties. Two liters of sediment were thus sampled from the sieved sediments resulting from the excavation of layers H and I (Fig. 2). Layer H represents well the major sedimentary matrices that characterize the sequence, with sandy-silty deposits containing pebbles and gravels. Subsequently, 45 experimental replicas were realized on the geological samples (Fig. 3) by an experienced knapper (E. C.) by direct percussion using a rounded flint nodule as a hammerstone.

The experimental pieces were realized considering: (i) the two different raw material qualities; (ii) the reduction method frequently observed in the archaeological record (for instance Levallois flake production, but also the shaping of a bifacial tool from a nodule); (iii) different products and end-products (tools, flakes, and cores); (iv) different morphometrical characteristics of the artifacts; and (v) differences in the presence and surface coverage of cortex (Table 1). All these parameters were chosen to create replicas that share similar characteristics with the archaeological materials. All the experimental pieces and the sampled sediment were sent to the Laboratory of Technological and Functional Analyses of Prehistoric Artefacts (LTFAPA) of Sapienza University of Rome, where a tumbling experiment took place. All the pieces were measured, weighed, and photographed. Proximal, mesial, and distal edge thicknesses were measured. Twenty-eight pieces were not subjected to the mechanical experiment, since they were kept for the chemical alteration experiment carried out at the PACEA laboratory, at the University of Bordeaux. On

Table 1 Characteristics of the experimental flint artifacts used in the tumbling experiments E1 and E2

id	Raw material	Typology	Cortex coverage [a]	Length (cm)	Width (cm)	Edge thickness (cm)	Weight (g)	Experiment
1	Coarse-grained flint	Biface	1	9.5	6.1	3.1	196.7	E1
3	Coarse-grained flint	Flake	2	5.5	3	0.6	12	E1
4	Coarse-grained flint	Flake	2	3.2	2.2	0.7	4.9	E1
5	Coarse-grained flint	Flake	1	3.8	2	0.3	2.5	E2
8	Coarse-grained flint	Flake	2	8.7	9.6	3.2	228.9	E1
9	Coarse-grained flint	Flake	1	6.8	6.9	2.3	82.6	E2
10	Coarse-grained flint	Flake	1	6.2	3	0.8	17	E1
15	Coarse-grained flint	Flake	1	7	6.4	2.4	102.5	E1
18	Coarse-grained flint	Flake	2	9.9	4.8	1.9	96.9	E2
22	Fine-grained flint	Levallois core	1	6	5.5	3.1	88.9	E1
23	Fine-grained flint	Levallois flake	1	6.1	4.1	0.8	17.9	E1
25	Fine-grained flint	Flake	1	3.9	4.5	0.8	10.7	E1
29	Fine-grained flint	Flake	3	3	1.8	0.4	2.7	E1
31	Fine-grained flint	Flake	3	2.8	2.6	0.8	6.6	E1
33	Fine-grained flint	Flake	1	6.6	4.8	1.7	40.7	E1
37	Fine-grained flint	Flake	1	3.2	2.8	0.4	3.3	E2
42	Fine-grained flint	Flake	2	7	3.4	1.8	28.4	E2

Table 1 (continued)

[a]Cortex coverage: 1 (<20% of the whole surface); 2 (20–30% of the whole surface); 3 (>30% of the whole surface)

the 17 remaining pieces, we used a two-component silicon (Vestige Light-Fast Trayart) to obtain molds of edges and ridges, which provide an exact copy of the artifact's fresh surfaces (a technique already used in, *e.g.*, Asryan *et al.*, 2014; Lemorini *et al.*, 2019; Ollé & Vergès, 2014).

The tumbling experiments

To evaluate the effects of flint artifacts transported in a hydraulic environment, experimental flint replicas were introduced in a tumbling machine (Lortone Mod. 3 A 001–092, 220 V with a rotative speed of 48 rpm) that simulates transport and rolling and produces impacts caused by small gravels and particles' saltation during water flow (Fig. 4) (for similar experiments, Burroni *et al.*, 2002; Bustos-Pérez *et al.*, 2019; Bustos-Pérez & Ollé, 2024a; Chu, 2016; Galland *et al.*, 2025; Levi-Sala, 1986; Mazzucco *et al.*, 2013; Shackley, 1974; Schoville *et al.*, 2016). We calculated artifact density from layer H (90 artifacts/m^3) and reported it with the volumetric capacity of the tumbler's barrel (0.8l) as follows: 1 m^3:90 = 0.0008 m^3: X, where $X = 0.072$ artifacts in 0.0008 m^3.

Therefore, we decided to insert a single artifact at a time in the tumbling machine to respect the low density of artifacts recorded in layer H. Twelve replicas were individually put in the barrel with 50% sediment and 50% water (Fig. 4). We refer to this experiment as experiment 1 (E1). The tumbling machine was then activated during three rounds of 5 h each. At the end of each round, pieces were extracted and cleaned individually with industrial pure (99.9%) acetone and observed macroscopically and mesoscopically. Before the microscopic observation, the artifacts were washed and placed in an ultrasonic tank containing demineralized water for 5 min. They were then extracted and rinsed with demineralized water. We decided to set up a sequential experiment to account for time as a potential variable and to observe gradual changes that eventually occurred on the artifact's surfaces, that is, to document the development of the alterations (*e.g.*, Ollé & Vergès, 2014). For the mesoscopic and microscopic observations, we used a stereomicroscope Nikon SMZ 745 T (oculars × 10, objective × 1, magnification range × 0.67X–× 6.7) and a Nikon Eclipse 600 (oculars × 10, objectives × 5 and × 10) coupled with an optical fiber reflected light system. Photographs were taken using a Digital Camera Nikon DXM1200 and a Digital Camera OPTIKA. Also, after each round of tumbling, molds were realized using a two-component silicon (Vestige Light-Fast Trayart), to provide an exact copy of the edges and to document the sequential alteration for further analysis, if needed. For this study, all the observations and measurements (including those on the chemically altered artifacts) were made on the lithics' surfaces.

To evaluate a hypothetical situation of a higher density of artifacts transported, a group of 5 specimens was introduced simultaneously into the tumbling machine with 50% sediment and 50% water. We refer to this experiment

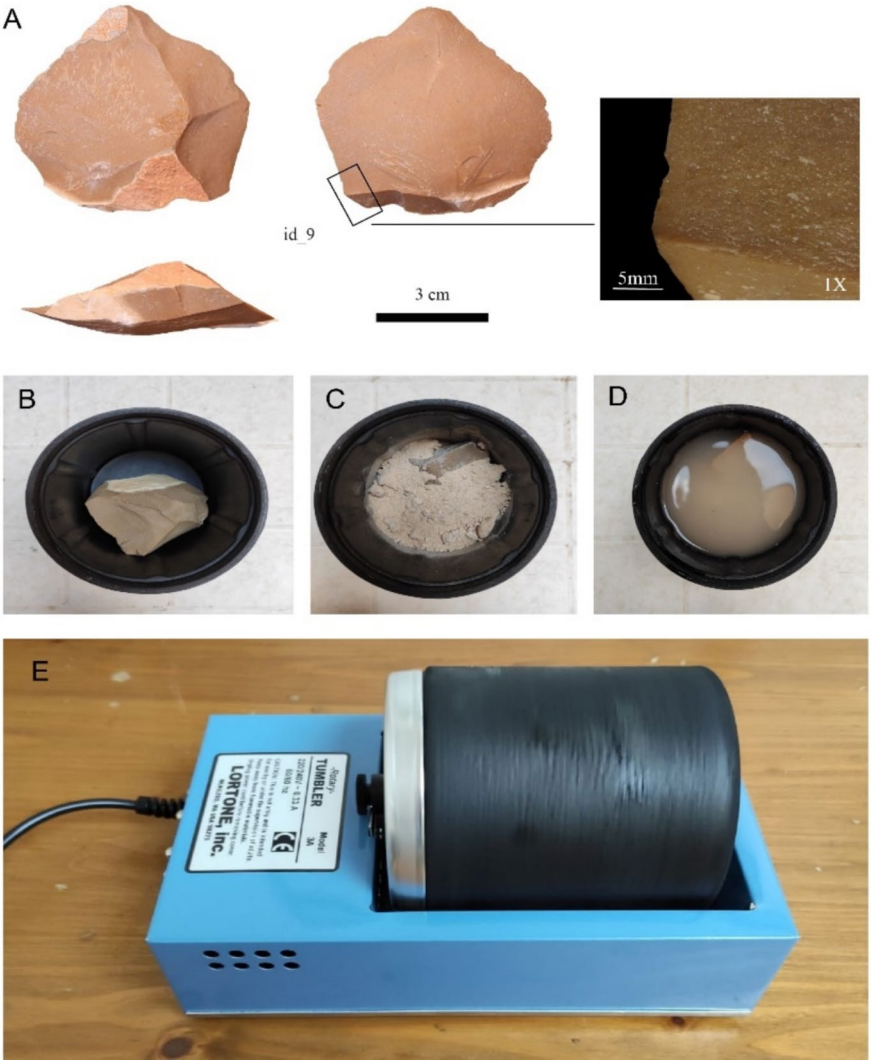


Fig. 4 The tumbling experiment with (**A**) photographic documentation of the fresh state of the artifact replicas (sharp outlines and absence of scars); (**B**) their introduction into the tumbling barrel (specimen id_1 in the picture); (**C**) sediment and (**D**) water filling; (**E**) activation of the rotating tumbler during three rounds of 5 h each

as experiment 2 (E2) (Table 1). After each round of 5 h of tumbling, for a total of 15 h, pieces were extracted, cleaned, observed, and documented macroscopically and mesoscopically, as for E1.

The chemical experiment

The chemical experiment was set to evaluate the degree of patina formation over the two different types of flint present in the 16/15 lithic assemblage. Of the 28 pieces not used in the tumbling experiment, we selected 8 flakes made of the two different flint varieties (coarse-medium-grained and fine-grained) knapped out from five different raw flint blocks. The remaining twenty pieces were kept as control samples and for further experiments. The eight flakes were placed in different plastic boxes containing a sodium hydroxide basic solution (NaOH, 5 mol/l), a potassium hydroxide basic solution (KOH 5 mol/l), and an acid solution of hydrochloric acid (HCl) at 37% concentration (Table 2). We thus followed the methodology proposed by Caux and colleagues (2018), setting up a sequential experiment by placing the artifacts for 8 days in a stove set at 50 °C. On days 1, 4, 6, and 8, we removed the flakes to record the degree of patina formation at each stage. Two specimens (id_20 and id_45) were split into three portions and altered using the three different solutions (Table 2) to avoid differences in microtopography and microporosity that may have been present in the different raw blocks of flint. Before starting the experiment and at the end of each sequential day, we (i) realized silicone molds using a two-component silicon (PRESIDENT microSystem Polyvinylsiloxane); (ii) documented macroscopic changing in color and surface with a Leica Z16 APO macroscope equipped with a camera (FLEXACAM C); and (iii) measured changing of surface roughness by microtopography measurements and 3D scanning of only ventral surfaces using a confocal microscope Sensofar S-Neox. For the confocal measurements, we measured far from the edges random surfaces of 820 × 620 µm with the Nikon × 20 objective (WD = 4.5 mm, NA = 0.45) on each specimen. We thus repeated the measurements upon the same surfaces on days 0, 1, 4, 6, and 8. Following Caux *et al.* (2018), we used the SensoMAP Premium 8.1 software to remove the outliers resulting from measurement errors and to fill the non-measured points, and chose a Gaussian filter of 80 µm cut-off.

Table 2 Characteristics of the experimental flint artifacts and chemical solutions used in the experiment

id	Raw material	Typology	Cortex coverage [a]	NaOH	KOH	HCl
14	Coarse-grained flint	Flake	1	X		
34	Fine-grained flint	Flake	1	X		
6	Coarse-grained flint	Flake	1		X	
24	Fine-grained flint	Flake	1		X	
11	Coarse-grained flint	Flake	2			X
30	Fine-grained flint	Flake	3			X
20	Coarse-grained flint	Flake	3	X	X	X
45	Fine-grained flint	Flake	3	X	X	X

[a]Cortex coverage: 1 (0–20% of the whole surface), 2 (20–30% of the whole surface), 3 (> 30% of the whole surface)

To gain better detail and to account for surface heterogeneity, each surface was divided into four equal parts, which were measured. After testing different roughness parameters, we chose the square root mean height (Sq), the height distribution (Ssk), and the void volume (Vv) parameters to characterize and document sequential changes in surface microtopography and microporosity through the experiment (*e.g.*, Caux *et al.*, 2018; Coco *et al.*, 2025; Galland *et al.*, 2019). Sq (μm) characterize microtopography and it is the calculated standard deviation of the height distribution of a surface, obtained by squaring each height value in the dataset, then taking the square root of the mean; the Ssk is the skewness of the height distribution calculated by the measure of the asymmetry of the surface deviation about the mean plane; Vv ($\mu m^3/\mu m^2$) characterize microporosity and it is a measure of the void volume provided by the surface between various heights (*e.g.*, Blunt & Jiang, 2003). All results not provided here, as well as the underlying data, programming code, and the figures of the alteration presented in the text, are available in an online repository at https://doi.org/10.5281/zenodo.17076226.

PDSM attributes definition

The description of the PDSM attributes observed in our samples (edge/ridge rounding, fractures, micro-flaking, polish, bright spot) is based on prior use-wear and PDSM research (Asryan *et al.*, 2014; Burroni *et al.*, 2002; Bustos-Pérez *et al.*, 2019; Levi-Sala, 1986; Mazzucco *et al.*, 2013; Prost, 1993; Tringham *et al.*, 1974) and includes the type of wear, its morphology, position, distribution, and orientation.

Among the mechanical PDSM, we identified:

Edge rounding, which corresponds to macro- and mesoscopic abrasion of the surface. This abrasion results from mechanical friction during particle and artifact transport and typically begins with micro-chipping of the edges, subsequently developing into a progressive "grinding down of the ridge" (Shackley, 1974). The outcome is a gradual smoothing and rounding of the edge, which appears dull.

Ridge rounding, which may result in an increase in ridge width. The measurement of ridge width can be used as a proxy to define and quantify degrees of alteration.

Based on the overall appearance of edges and ridges, we classified the rounding as follows: fresh, low, medium, and high. In this study, only qualitative characteristics of rounding were recorded (*e.g.*, Santonja & Pérez-González, 2001), although quantitative approaches to record degrees of rounding have been proposed elsewhere (*e.g.*, Bustos-Pérez *et al.*, 2019; Chambers, 2016; Shackley, 1974).

Microfractures/Microflaking correspond to macro- and microscopic scars produced by impact or crushing forces exerted over the artifact's edges during transportation. Such bending or percussive forces can fracture edges or detach microflakes, resulting in characteristic edge damage (Hosfield & Chambers, 2005; Tringham *et al.*, 1974). In this study, microflake morphology and ter-

mination (feather, step, hinge, snap) have been recorded using Fischer *et al.* (1984) descriptions.

Polish refers to microscale surface abrasion generated by particle friction on the artifact's surface. It results in an alteration of the microtopography, producing a dulling effect of the surface, which often appears bright (Bustos-Pérez & Ollé, 2024a, 2024b; Grace, 1996; Levi-Sala, 1986).

Bright spots are isolated, microscopic reflective spots of abrasion that result from the friction between artifacts and particles during transportation. They are generally located on the more elevated topographic areas of the surfaces (Levi-Sala, 1996; Mazzucco, 2014; Van Gijn, 1989).

We also observed the removal of small chips of material from the artifact's surfaces. In contrast to microflaking, these removals do not leave characteristic scars resulting from conchoidal or bending forces. Instead, they appear to result from surface exfoliation processes.

Among the chemical alterations, we observed:

The development of the *white patina*, a process of dissolution, integration, or deposition of silica that primarily occurs under elevated pH conditions in basic soils (Rottländer, 1975; Thiry *et al.*, 2014). The white patina formation, which is linked to the specific environmental context (Burroni *et al.*, 2002), can substantially modify the surface microtopography and microporosity of the lithic artifacts (Caux *et al.*, 2018; Galland *et al.*, 2019).

Archaeological sample

A sample of 11 archaeological artifacts recovered from site 16/15 was examined both macro- and microscopically to detect the presence (if any) of mechanical and chemical PDSM. The sample was selected to encompass the variability documented in the archaeological record. The selection criteria considered the following factors: differences in raw material (brownish coarse-grained flint, brownish medium-to-fine-grained flint, darkish very fine-grained flint); differences in products and end-products (tools, flakes, and cores); differences in the morphometrical characteristics of the artifacts; and differences in the stratigraphic distribution of the artifacts. One artifact comes from layer B, one from layer K, three from layer G, three from layer H, and three from layer I (Table 3). While this limited sample may not be representative of the whole taphonomic processes at play at the site, it serves as a preliminary and exploratory test case. Ongoing analysis on a much larger sample will provide additional information. The artifacts were cleaned individually with industrial pure (99.9%) acetone and analyzed microscopically using a portable microscope (Dino-Lite AM7013MZT) at × 20X and × 30 magnification at the Institut National du Patrimoine in Tunis. Photographs were taken, and alterations were measured with the DinoCapture Software.

Table 3 Characteristics of the sample of archaeological flint artifacts from site 16/15

id	Raw material	Patina	Typology	Layer	Cortex coverage (%)	Length (cm)	Width (cm)	Thickness (cm)	Weight (g)
T1	Coarse-grained flint	Extensive	Point	H	0–20%	5.8	4.1	1.1	28.5
T2	Medium to fine-grained flint	Light	Levallois flake	G	0–20%	7.3	6.9	1.6	80.3
T3	Fine-grained flint	No	Flake	H	0–20%	1.9	3	1.4	8
T4	Coarse-grained flint	Extensive	Flake	I	20–30%	15.1	7.3	2.3	284.6
T5	Coarse-grained flint	Extensive	Flake	G	0–20%	6.1	6	2.2	74.4
T6	Coarse-grained flint	Extensive	Core	I	0–20%	6.9	5.6	2.8	94.4
T7	Coarse-grained flint	Extensive	Biface	K	0–20%	9.8	7.9	3.2	251.1
T8	Coarse-grained flint	Extensive	Levallois flake	I	0–20%	7.7	5	1.3	67.2
T9	Fine-grained flint	Light	Flake	B	20–30%	1.7	1.1	1.2	2
T10	Medium to fine-grained flint	Light	Flake	G	0–20%	2.9	5	8	12.9
T11	Coarse-grained flint	Extensive	Backed piece	H	0–20%	2	3.1	7	4.7

Results

Tumbling experiment results

E1 after 5 h of tumbling

After 5 h of tumbling, only one specimen did not suffer from any post-depositional surface modification. All other specimens were altered, although only very light alterations occurred. The most prevalent PDSM consisted of the detachment of small, thin chips of material, documented on seven artifacts (58% of the assemblage). This alteration is more consistent with processes of surface exfoliation rather than with the action of percussive forces (Fig. 5, bottom). Microflaking was observed on 6 artifacts (50%), followed by slight edge rounding (3; 25%), edge fracture (2; 17%), and concretions dissolution (1; 8%) (Fig. 6A). No polish formation was observed. On 8 out of the 12 artifacts (67%), we register the cooccurrence of more than two alterations. Their occurrence is predominant on the fine-grained artifacts over which we recorded 18 alterations, compared to 10 on the coarse-grained ones. On artifact id_23, we observed the simultaneous development of four alterations of three different typologies (removal of thin chips of

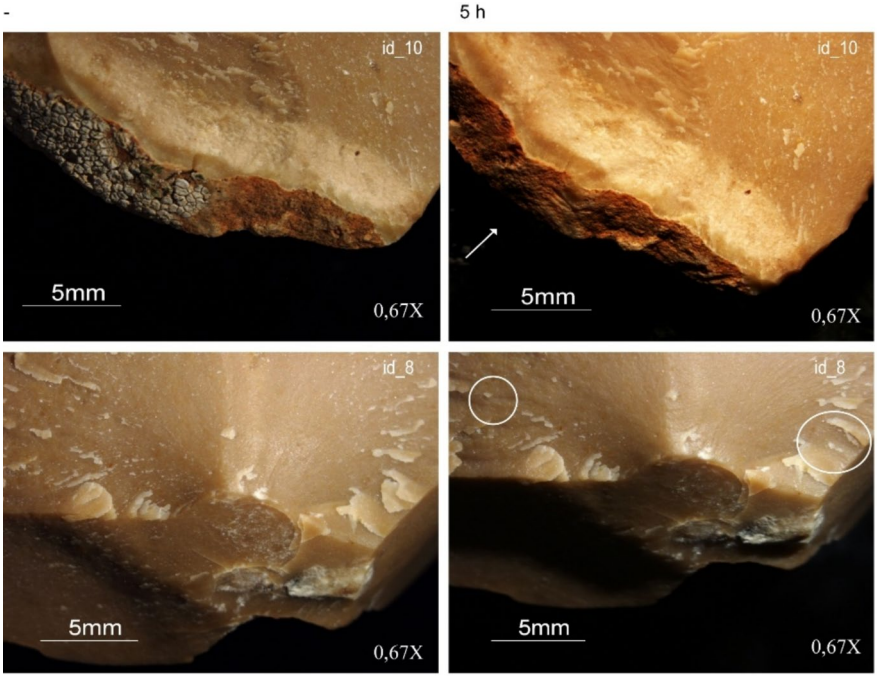


Fig. 5 Photographs showing two alterations observed frequently after the first round of tumbling (5 h) (E1). Top: the concretions on the id_10 cortical surface have disappeared. Bottom: removal of small and thin chips of material observed over some portions of artifact id_8

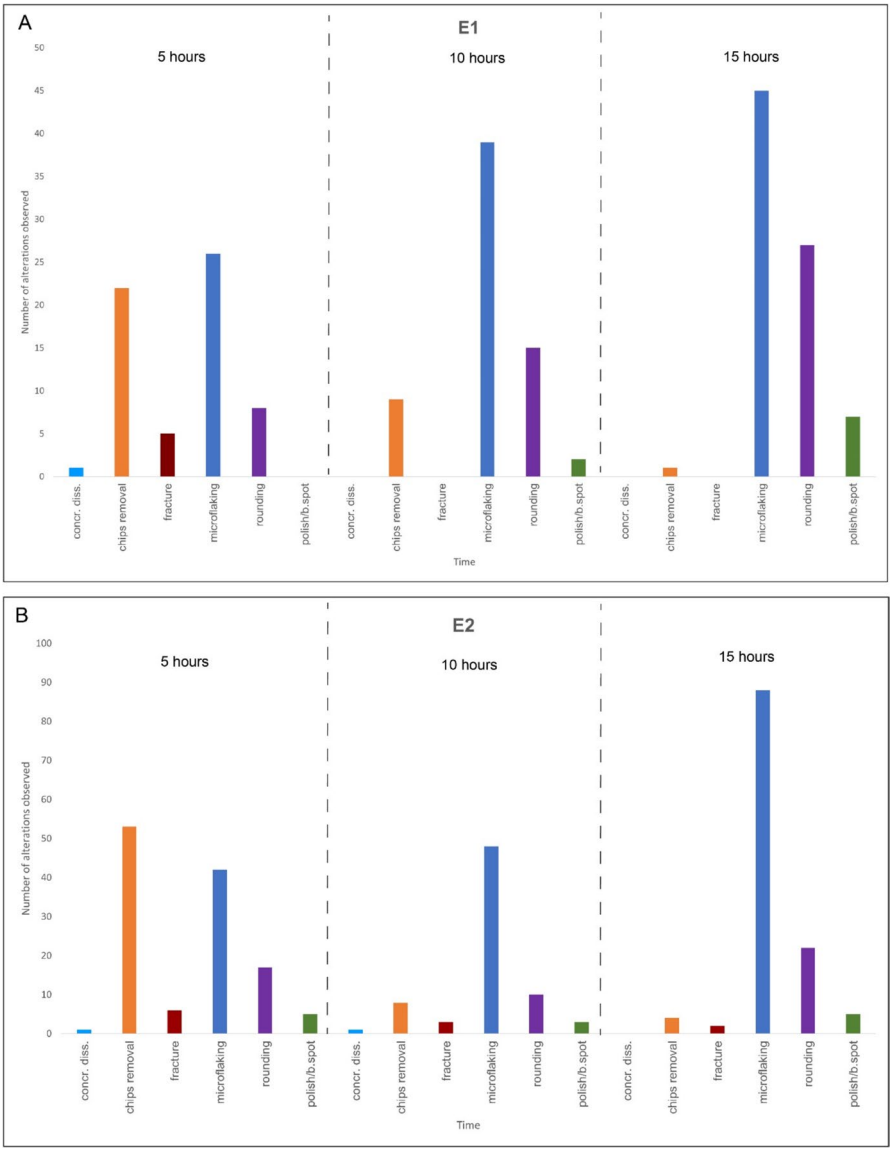


Fig. 6 Amount, increase, and decrease of PDSM as recorded after 5, 10, and 15 h of tumbling during **A** experiment 1 (E1) and **B** experiment 2 (E2)

material, microflaking, fracture), equally distributed on both the dorsal and the ventral surfaces. Microflaking and edge rounding, two common PDSM observed in tumbling experiments (Asryan *et al.*, 2014; Burroni *et al.*, 2002), were most frequent on fine-grained artifacts (Fig. 7). Rounding is slight, never extending more than 1–2 mm from the edges. Microflaking scars are semi-circular with hinge terminations and measure 0.5–1 mm in length.

Fig. 7 Photographs showing the sequential development of alterations along some artifacts' edges after ▶ each round of tumbling (E1). In detail, the progressive development of edge rounding is visible on id_31 and id_23; Microflaking (id_15; id_25; id_23) increases over time and produces scars with different characteristics depending on the raw material (see text for details)

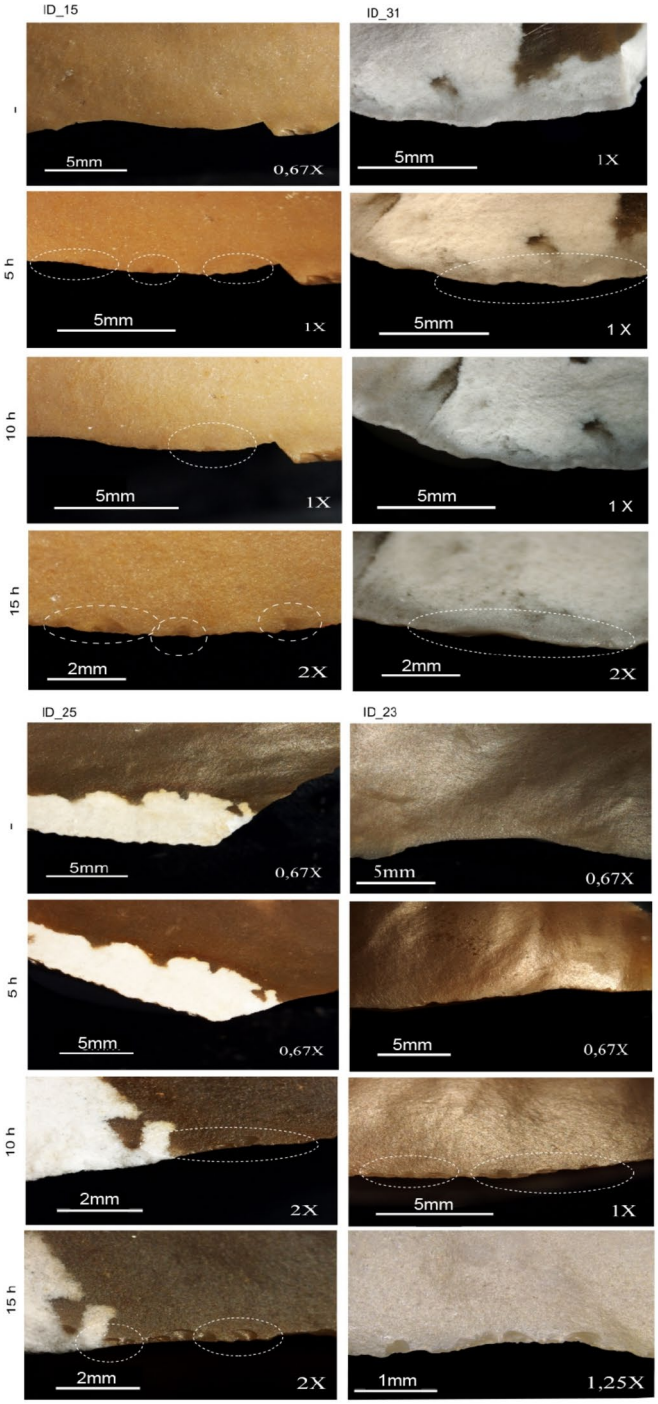
Overall, scars are mostly distributed on the ventral surfaces and distal portions, mainly affecting the thinner edges of the artifacts, regardless of flint variety.

E1 after 10 h of tumbling

After the second stage of the experiment (10 h of tumbling), the entire assemblage exhibited clear evidence of PDSM. The removal of thin chips of material was observed on three artifacts (25% of the whole assemblage), microflaking on 12 (100%), and edge rounding on 8 (67%) (Fig. 6A). During this second stage of the experiment, 11 artifacts (92%) developed more than two alterations, and specimen id_33 exhibited the co-occurrence of five alterations. Again, alterations are particularly intensive on artifacts made of fine-grained flint. Rounding becomes increasingly pronounced, with edge rounding extending between 1 and 2.5 mm in length (Fig. 7). Surfaces appear shiny even at low magnifications, with smoothing and incipient polish affecting the highest point of the edges (Fig. 8). Microflaking is well developed and affects longer portions of the edges. In some cases, scars start grouping and coalescing along the edges (Fig. 7). Scars have semi-circular and half-moon morphologies with hinge and feather terminations (as also observed by, *e.g.*, Mazzucco, 2014), and range between 0.5 and 1 mm. No damage was observed on the cortical portions or on the ridges of the artifacts. PDSM developed both on ventral and dorsal surfaces, with a slight predominance on the former. Overall, damage mainly affects the thinner outlines of the artifacts, although on the fine-grained flint artifacts, it is distributed more evenly on both thick and thin edges. Alterations are consistently distributed on the distal portions, but we noticed an increase in damage distribution on proximal, as well as a slight but significant increase on mesial portions.

E1 after 15 h of tumbling

At the end of the third round of tumbling, PDSM developed on the whole assemblage. The removal of thin chips of material was observed on one artifact (8% of the whole assemblage), microflaking on 11 (92%), and edge rounding on 11 (92%) (Fig. 6A). Specimens id_10 and id_8, two coarse-grained flint flakes, have not developed microflaking and rounding, respectively. All artifacts displayed more than two alteration types, with fine-grained flint specimens being the most affected. Rounding ranges between 2 and 5 mm in length and affects the artifacts' edges, which appear smoother and shinier. Furthermore, rounding was detected on the cortical edges of four fine-grained flint artifacts. Rounding and abrasion, which have also started to form along some cortical edges (id_31) and ridges (id_22), heavily smooth the surface topography, except for the depressions that remain unaffected by alteration


ID_15
ID_31
-
5mm
0,67X
5mm
1X
5 h
5mm
1X
5mm
1 X
10 h
5mm
1X
5mm
1 X
15 h
2mm
2X
2mm
2X
ID_25
ID_23
-
5mm
0,67X
5mm
0,67X
5 h
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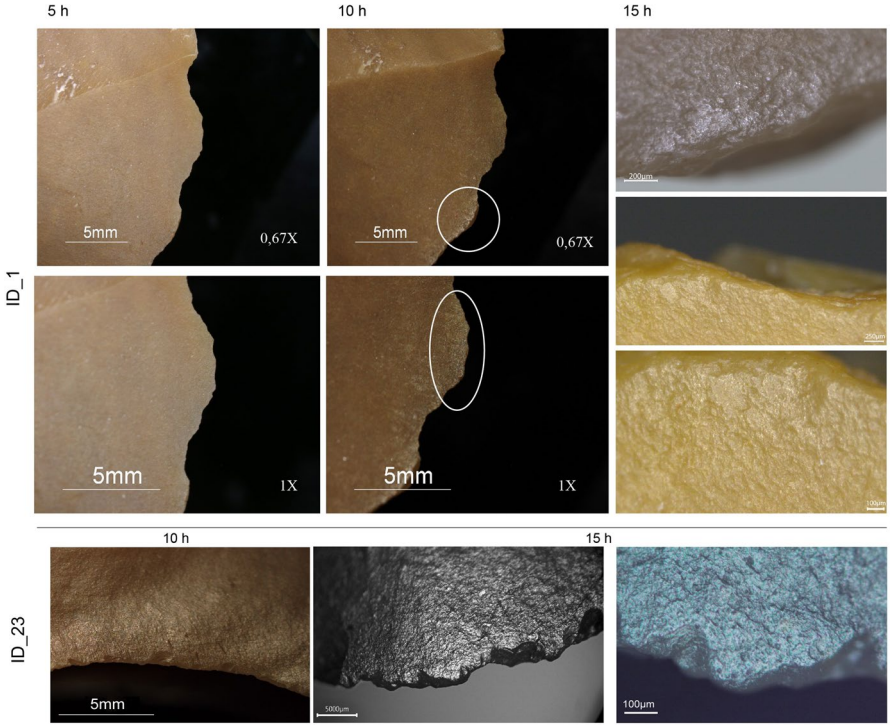

Fig. 8 Edge rounding results in heavy abrasion and smoothing of the artifacts' surfaces and microtopography. The rounded edges present a shiny appearance even at low magnifications, with an incipient development of polish along the edges (id_1 and id_23). Although rounding starts to develop already after 5 h of tumbling, polish was more appreciable after 10 h and increased after 15 h

(Fig. 9). On the same ridge of specimen id_22, we also observed an incipient formation of microflaking. Microflaking is well developed. Nevertheless, scars are never homogeneous in size, nor equally oriented (Fig. 7). Microflaking scars range between 0.5 and 2.5 mm in length and are often overlapping over scars that have previously formed (during the first two rounds of the experiment), producing a widening of existing damage, and coalescing (Fig. 7). Polish and isolated bright spots (Levi-Sala, 1986) also developed on the protruding portion of the artifacts (Fig. 9). Alterations occurred on both surfaces, with a slight predominance over the ventral one. Thick and thin edges were affected by the development of PDSM, along the distal, the proximal, and, to a lesser extent, the mesial portions of the artifacts.

E2 after 5 h of tumbling

After the first 5 h of tumbling, all specimens were altered. Concretion dissolution was recorded on one specimen (20% of the whole assemblage), the removal of thin chips of material on three (60%), edge fracture on two (40%), while microflaking and rounding affected the whole assemblage (Fig. 6B). Each artifact developed more than two

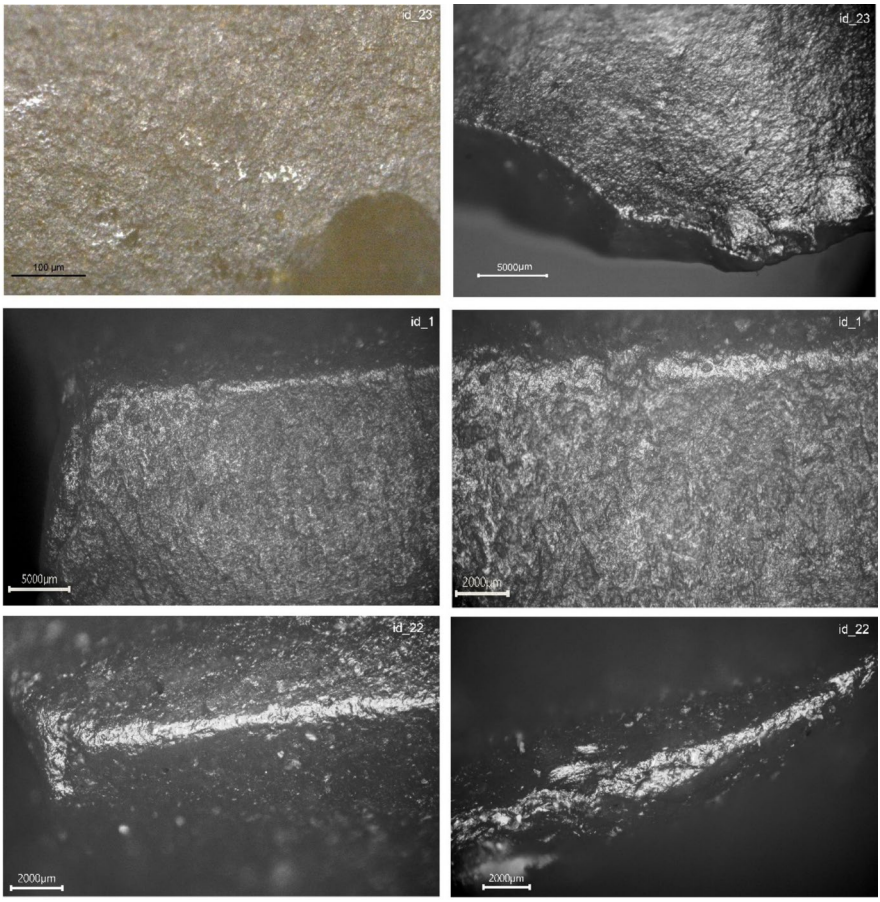

Fig. 9 Bright spots (id_23), rounding, and reflective polish (id_23, id_1, and id_22) develop along the artifact's edges, ridges, and surfaces after 15 h of tumbling

types of alteration, with a slight predominance of damage recorded on the fine-grained flints. Rounding, although light and with limited extension (2–5 mm in length, 1–2 mm from the edges inwards), was observed along the edges, particularly on the more prominent surfaces. Microflaking was more intense and extensive compared to E1 (first round) (Fig. 10). We recorded semi-circular, half-moon, and cone-initiated scars ranging between 1 and 5 mm in length. On the finer variety of flint, they tend to be grouped, whereas on the coarser-grained flint, they are mostly isolated. Moreover, they vary both in size and orientation. Incipient polish was observed on specimens id_18 and id_37 (Fig. 11). No cortical portions were affected by the development of wear, although slight ridge rounding was documented on artifact id_9. Both the ventral and the dorsal surfaces were evenly affected by the development of PDSM, which mostly forms on the thinner edges. Finally, the PDSM distribution is greater on the distal and mesial portions of the artifacts than on the proximal ones.

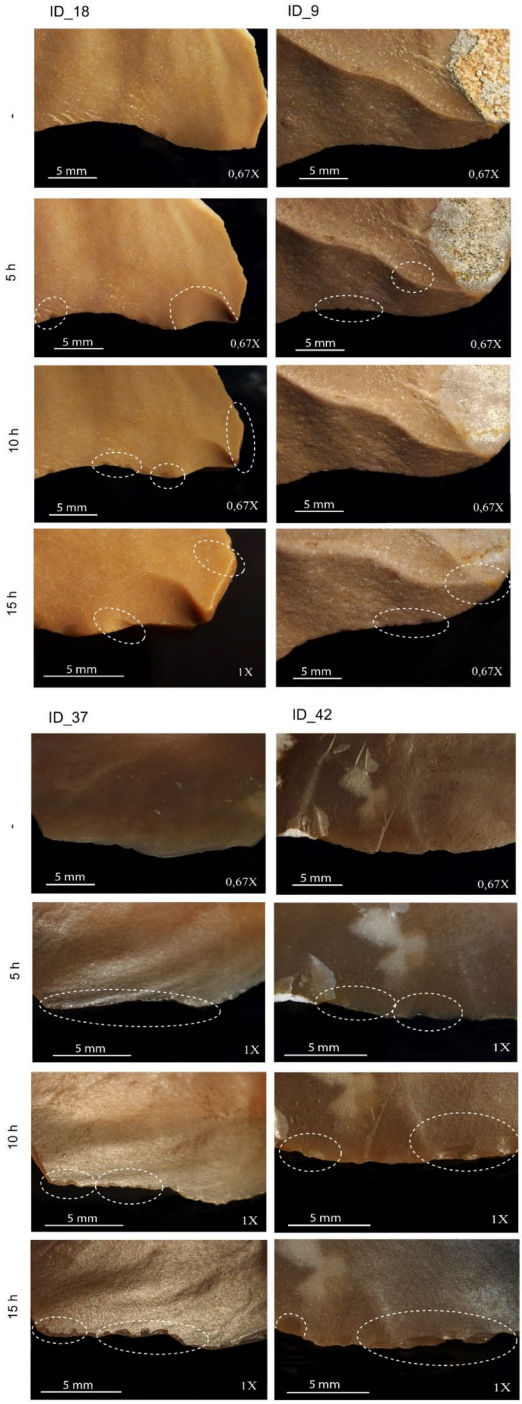

Fig. 10 Photographs showing the sequential development of alterations after each round of tumbling (E2). In detail, the progressive development of microflaking and edge rounding is visible on all artifacts; concretion dissolution, removal of thin chips, edge and ridge rounding on id_9; polish on id_18 (see text for details)

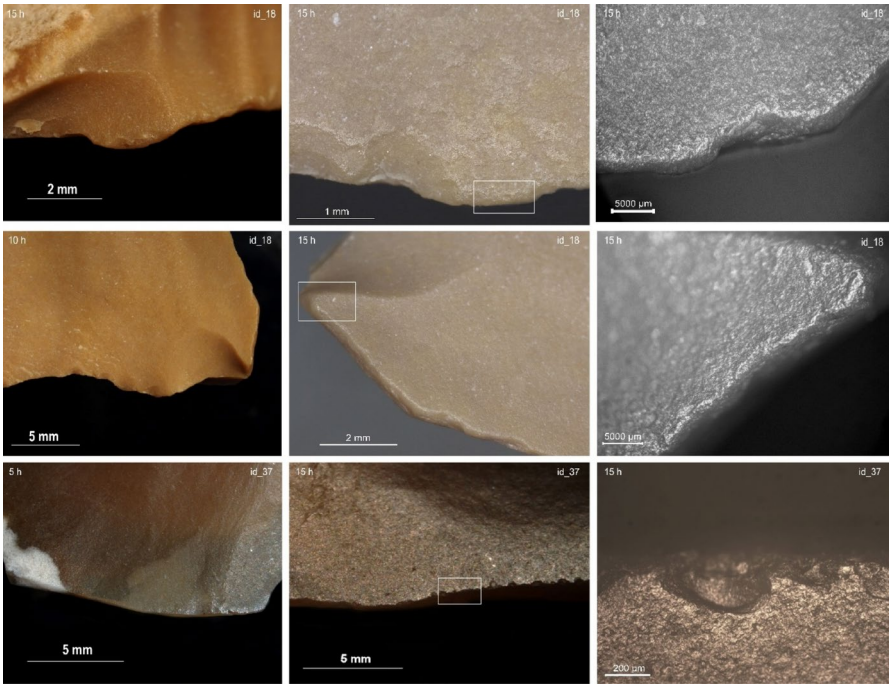

Fig. 11 Development of polish and rounding on elevated surfaces (id_18 and id_37) and bright spots (id_37) at different stages of the experiment (E2)

E2 after 10 h of tumbling

All specimens were altered after the second round of tumbling. The removal of thin chips of material was observed on one artifact (20% of the whole assemblage), edge fracture on one (20%), and microflaking and edge rounding on the whole assemblage (Fig. 6B). All the artifacts developed more than two types of alterations, with a predominance of damage recorded on the fine-grained ones, on which alterations are intensive and well developed (Fig. 10). Rounding is barely perceptible and restricted only to some prominent surfaces, maybe due to the stabilization of the edges and a smoothing steady phase, as recorded by Burroni *et al.* (2002). Conversely, the development of microflaking was severe, sometimes continuously distributed along the edges, except on the coarse-grained artifacts, where the scars are isolated. Scars are semi-circular, triangular, and cone-initiated, spanning between 1 and 4 mm in length, with hinge and feather terminations. They vary both in terms of size and orientation. No PDSM was recorded on the ridges and cortical portions. They instead evenly affected both the ventral and the dorsal surfaces, the thicker and thinner edges, with a slight prevalence of the latter. Moreover, PDSM are distributed on the distal, mesial, and proximal portions of the artifacts.

E2 after 15 h of tumbling

At the end of the third round, PDSM developed intensively on the whole assemblage. The removal of thin chips of material was observed on one artifact (20% of the whole assemblage), edge fracture on one (20%), and microflaking and edge rounding on the whole assemblage (Fig. 6B). Again, the fine-grained flint artifacts were more affected by PDSM. Rounding and smoothing affected the edges and ridges extending between 2 and 5 mm in length and 1 and 2 mm from the edges inwards. On id_5, we observed the rounding of a cortical edge. Microflaking is well developed along the edges, with semi-circular, triangular, half-moon, and cone-initiated morphologies and hinge and feather terminations. They span between 1 and 3 mm in length. On the fine-grained flint artifacts, we observed the grouping and coalescing of the scars (Fig. 10). Polish also expands on prominent surfaces (Fig. 11). PDSM formed on both surfaces, with slightly more incidence on the dorsal ones. They are distributed on thick and thinner edges, with prevalence on the latter, and on the distal, mesial, and proximal portions of the artifacts.

Chemical experiment results

The macroscopic observations and surface roughness measurements show an overall incrementation in patina formation over time (Fig. 12). However, differences exist and are related to both the chemical solution used and differences in the raw material texture.

On day 1, with the KOH solution, a change in color and the formation of a thin white patina distributed across the entire artifact surface were visible on the coarse-grained artifact, whereas no color change nor patina formation was observed on the fine-grained artifact. With the NaOH solution, a more extensive diffusion of the white patina was observed on the coarse-grained artifact compared to the KOH solution, and, similarly to KOH, it is more intense than on the fine-grained flint. Interestingly, some features (such as fossils and inclusions) became more visible due to the slight bleaching of the artifact surface. With the HCl acid solution, alteration began to develop on the cortical portions, which are probably partly calcareous, of both the coarse-grained and the fine-grained flints. On the flint surfaces, we observed only a slight change to a darker color and a loss in surface brightness.

On day 4, we observed a greater diffusion of the white patina and a loss of color and brightness on the experimental specimens treated with both the KOH and NaOH (Fig. 12B, F, J). However, the alteration is more intense and widespread with NaOH. Moreover, in both the KOH and NaOH experiments, the alteration has been more intense for the coarse-grained artifacts compared to the fine-grained ones. Although the white patina is now quite well developed, the original color of the flint surfaces is still visible on the artifacts. As per day 1, the bleaching of the artifact surface highlights fossils and inclusions (Fig. 12B), a phenomenon also observed by Michel and Rots (2022) during surface photodegradation. In contrast, the artifacts treated with the HCl solution exhibited mild changes (Fig. 12N). Once again, the alteration

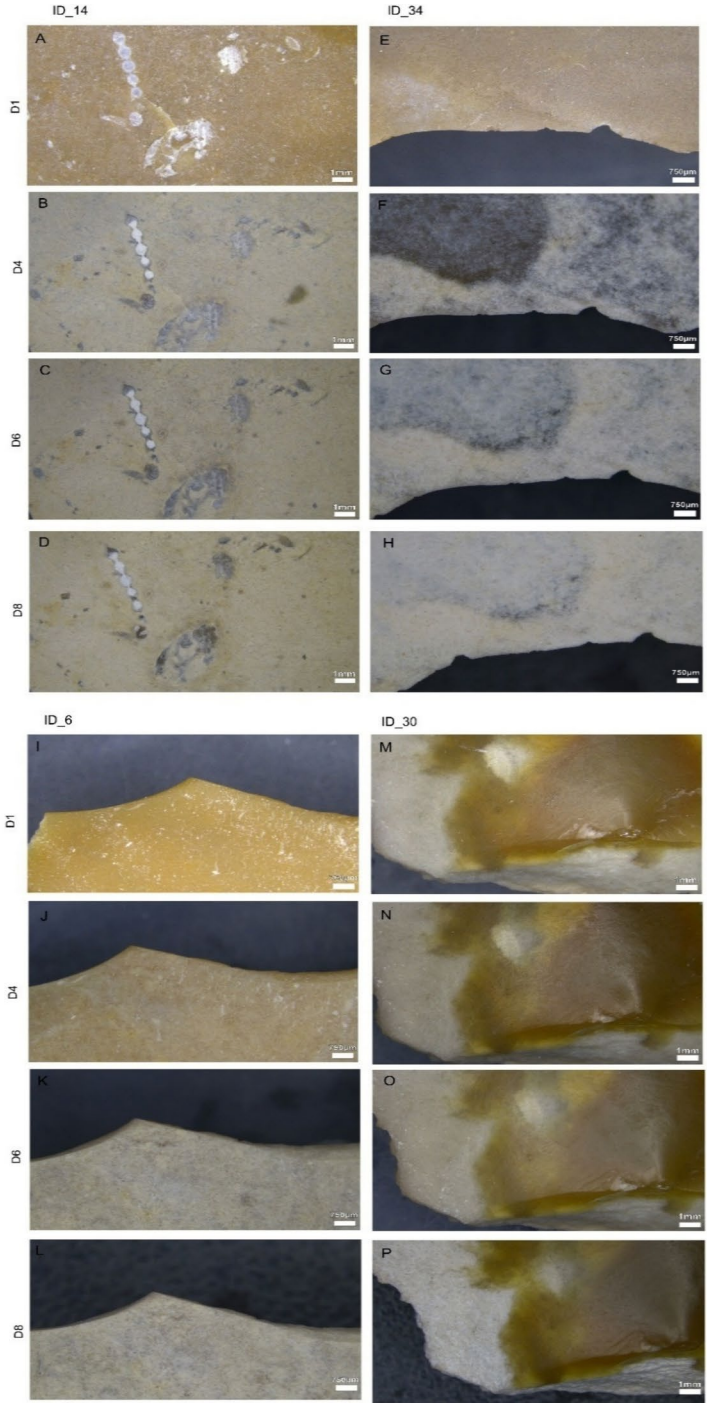

Fig. 12 Macroscopic observations of patina formation and changes in surface color over time according to different raw material qualities and chemical solutions. Id_14 and id_34, NaOH; id_6, KOH; id_30, HCl

predominantly affects the cortical portions, while on the flint surfaces, a spotted patina (which is lighter and thinner than the KOH and NaOH-induced ones) starts to be visible, particularly on the coarse-grained artifact.

On day 6, the specimens treated with the KOH solution show a widespread diffusion of the white patina over the whole surface. Nevertheless, the original color of the flints is still visible in many parts since the patina is sparsely distributed (Fig. 12K). The alteration remains more intense on the coarse-grained artifacts than on the fine-grained ones. The artifacts treated with NaOH show a widespread diffusion of a homogeneous, apparently thick white patina over the whole artifact's surfaces. The original color of the flint surfaces has radically changed, and it is almost unrecognizable (Fig. 12C, G). However, macroscopically, the changes that occurred between day 4 and day 6 appear moderate. This suggests that the patination process either decreases or stabilizes at this stage, or that the chemical solution may also be less active by now. The specimens treated with HCl exhibit pronounced alteration of the cortical parts, which appear both bleached and eroded simultaneously. By contrast, the core portions of the flint show no macroscopic signs of surface alteration nor color changes.

On day 8, the artifacts treated with the KOH solution show a slight increment in the diffusion of the white patina (Fig. 12L). A mild edge rounding has also been observed. The artifacts treated with the NaOH solution show a well-developed white patina over the whole artifact's surface. Features that were highlighted by the bleaching of the surface, like fossils and inclusions, are now also altered and less visible (Fig. 12D). Moreover, we observed rounding of the edges and ridges, and a loss of substance in the form of a small snap fracture and microflaking from artifact id_34, likely due to the weakening of the flint structure provoked by the chemical alteration (Fig. 12H). Finally, the artifacts treated with the HCl solution show an extensive development of a thick white patina, again exclusively over the cortical portions.

When we look at the surface roughness measurements, we observe more complex and variable processes and degrees of patina formation, which are influenced by both the applied chemical treatments and the compositional differences of the raw material.

On the fine-grained flint, the roughness of the surface is moderately affected by the KOH solution after 4 days and then remains almost stable over time. In contrast, with NaOH, there is a strong increase in the phenomenon over time, with significant variations in the measured roughness parameters (Fig. 13D, F). The chemical alteration generated by the NaOH solution significantly increases surface roughness (high Sq values and see Fig. 14C and D) over time by creating rare big voids (high Vv values and Fig. 14D) but remaining globally evenly distributed (Ssk values not changing and close to 0, Fig. 13E). This is related to the progressive development of the white patina (Fig. 12) by the dissolution of the siliceous material (Fig. 14D). By contrast, this fine-grained flint, with a high content of silica, is not affected so much by the acid solution, showing roughly stable Sq, Vv, or Ssk values over time.

On the coarse-grained flint, both the KOH and the NaOH basic solutions influence the surface texture, but with lower intensity than the fine-grained material. However, on the former, both the KOH and the NaOH solutions affect the variability of Sq values, resulting in a change of the measured property over time (Fig. 13A).

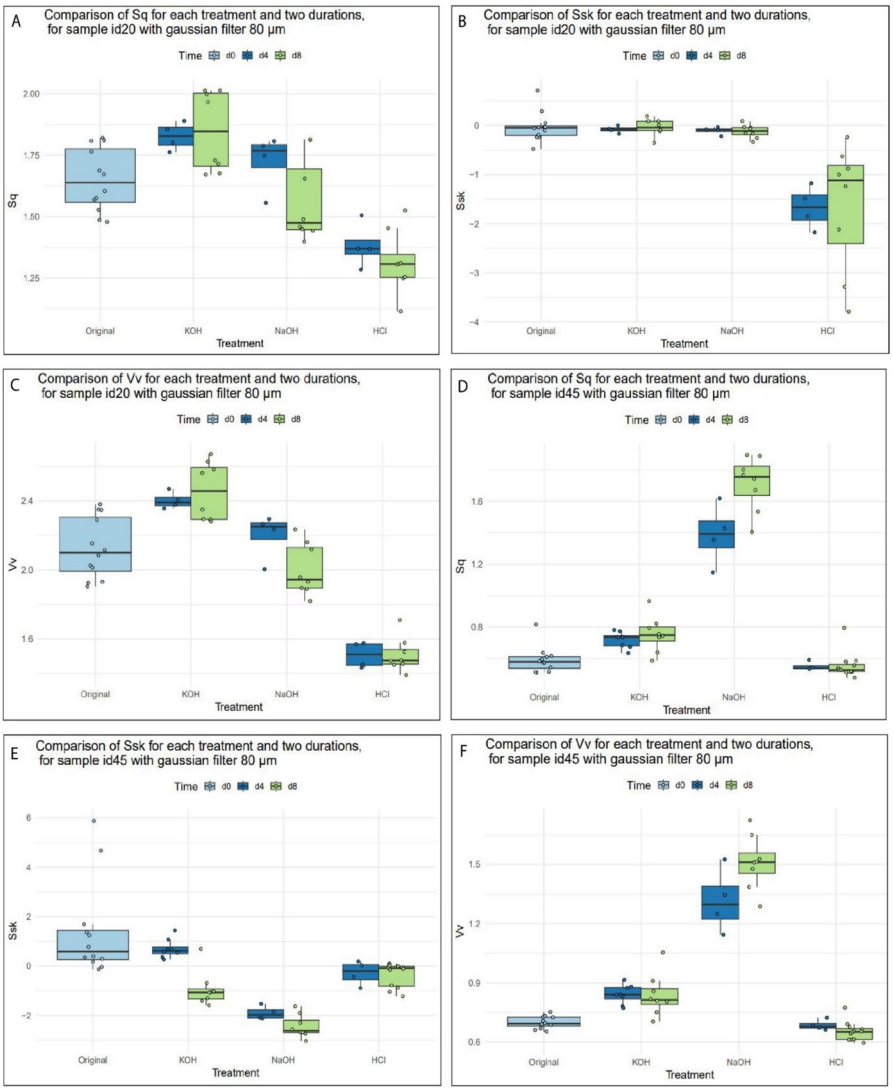

Fig. 13 Boxplot showing changes in Sq, Ssk, and Vv values over time for sample id_20 (coarse-grained flint) and id_45 (fine-grained flint) for each chemical treatment

KOH appears to increase the Sq values over time, with a slight increase until day 8, but no change in Ssk, suggesting a progressive increase in the difference in altitude of the measured points in both directions. This is indeed what is visible in Fig. 14A and B: the already pronounced topography of the coarse-grained flint at D0 is made even more pronounced, not only with deeper valleys but also keeping the peaks. NaOH also shows a slight increase in surface roughness until day 4 and perhaps a decrease and/or stabilization of Sq values until day 8, maybe indicating a stabilization of the process, as also observed macroscopically (that could also be related to

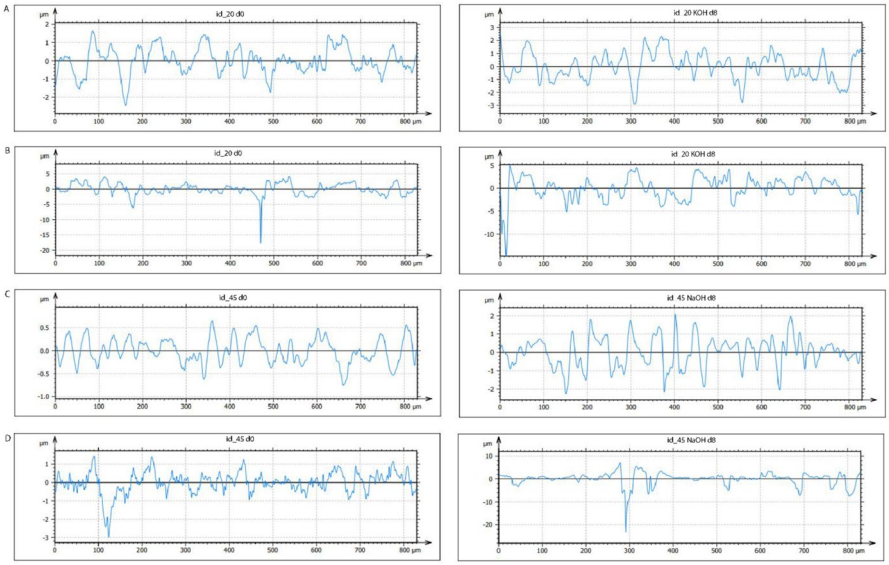

Fig. 14 Profiles illustrating changes in surface roughness from day 0 (left columns) to day 8 (right columns) of id_20 treated with KOH and id_45 treated with NaOH. Rows A and C are surface profiles measured on a line passing through the middle of the artifacts' surfaces. Rows B and D are surface profiles measured on a line passing through the lowest point of the artifacts' surfaces

a decrease in the pH of the basic solution itself). HCl shows low and stable Sq, Vv, and Ssk values, indicating that the peaks have been eliminated and the voids created quickly in this coarse and less silicified material. The HCl solution, while it does not influence the microtopography (Sq) nor the microporosity (Vv) values of the fine-grained flint so much, diminishes the roughness values of the coarse-grained ones.

These results indicate that while the fine-grained flints tend to alter almost exclusively within very elevated pH basic solutions and/or at a slower rate, the coarse-grained ones are affected by alteration within both low and high pH basic solutions, probably due to their more varied and less siliceous mineralogical composition.

Archaeological sample

All archaeological artifacts are affected by some degree of mechanical and chemical alteration, though in different proportions (Table 4). Macroscopically, the extent of chemical alteration is strictly related to differences in raw material quality. Coarse-grained flint artifacts present extensive surface alteration in the form of a widespread white patina, followed by the medium to fine-grained flint that presents a less intense and less diffuse thinner patina, and the fine-grained flint artifacts with light or no presence of patina (corresponding to group 1, group 2, and group 3, as defined by Cancellieri *et al.* (2022)) (Fig. 16). On the coarse-grained artifacts, the white patina is extremely developed and has affected the whole surface, including previous scars and cortical portions. Moreover, edges and surfaces are generally affected

Table 4 Type, distribution, location, and extent of PDSM identified on the archaeological sample from site 16/15

ID	Raw material	Patina	Layer	Microflaking scars (N)	Distribution [a]	Location [b]	Min Length (mm)	Max Length (mm)	Edge rounding	Ridge rounding	Polish
T1	Coarse-grained flint	Extensive	H	3	D/V	M/D	4.48	10.08	Medium	Medium	no
T2	Medium to fine-grained flint	Light	G	14	D/V	P/M/D	1.31	9.98	Low	Medium	yes
T3	Fine-grained flint	No	H	11	D/V	P/M	0.80	6.73	Low	Low	no
T4	Coarse-grained flint	Extensive	I	10	D/V	P/M/D	2.57	11.75	Medium	Medium	yes
T5	Coarse-grained flint	Extensive	G	13	D/V	P/M	3.44	12.26	High	High	no
T6	Coarse-grained flint	Extensive	I	10	D/V	P/M/D	1.78	13.13	Medium	Medium	no
T7	Coarse-grained flint	Extensive	K	5	D/V	M/D	2.06	9.96	High	High	no
T8	Coarse-grained flint	Extensive	I	13	D/V	P/M/D	3.09	10.02	High	High	no
T9	Fine-grained flint	Light	B	11	D/V	M/D	1.24	8.02	Low	Low	yes
T10	Medium to fine-grained flint	Light	G	19	D/V	P/M/D	0.80	10.80	Medium	Low	no
T11	Coarse-grained flint	Extensive	H	14	D/V	P/M/D	0.23	7.69	High	High	no

[a]Distribution: *D* dorsal, *V* ventral, *D/V* dorsal and ventral

[b]Location: *P* proximal, *M* mesial, *D* distal, *M/D* mesial and distal, *P/M/D* proximal, mesial, and distal, *etc*.

by intensive rounding, which is evenly distributed, likely suggesting that chemical alteration also contributed to the smoothing and rounding of the edges. Conversely, fine-grained artifacts display little to no patina, limited to a thin, weakly developed whitish film.

Mechanical PDSM, notably microflaking, and edge rounding, were consistently observed across the entire sample, regardless of raw material differences, stratigraphic provenance, or morphometric characteristics (Table 4; Fig. 15). Nevertheless, the fine-grained artifacts show higher proportions of observed microflaking scars (ratio of 13.75) compared to the coarse-grained ones (ratio of 9.71), suggesting that they are more prone to mechanical edge scarring. Conversely, edge rounding was recorded in higher proportions on the coarse-grained artifacts (ratio of 2.57), about twice that of the fine-grained ones (ratio of 1.25), regardless of stratigraphic provenance (Table 4). In addition, artifacts recovered from layers G, H, and I exhibit comparable percentages of recorded alterations, whereas those from layer B—and particularly from layer K—tend to display lower alteration frequencies (Fig. 15). Moreover, the relatively higher presence of microflaking recorded on artifacts from layer G may result from the larger number of fine-grained artifacts studied from this layer, given the overall sedimentary homogeneity between layers G, H, and I. This seems to confirm what was observed previously and in the experiment: that fine-grained flint is more prone to mechanical edge damage. However, these results should be interpreted with caution due to the limited size of the analyzed sample (particularly from layers B and K).

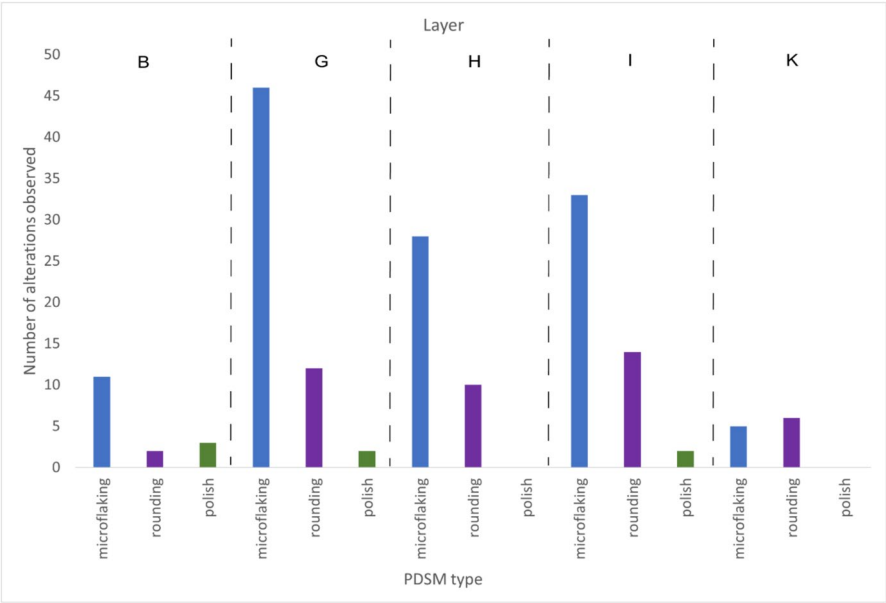

Fig. 15 Frequency of post-depositional surface modifications observed on the archaeological sample (microflaking, rounding, and polish), organized by artifacts' stratigraphic layer of provenance (**B**, **G**, **H**, **I**, **K**)

Microflaking scars are distributed along the edges of the artifacts, both on the ventral and the dorsal surfaces, with a slight predominance on the former. They range from 0.2 to 13 mm in length, and are semi-circular, triangular, and cone-initiated with mostly hinge and feather terminations. Considerable variability is observed in both their size and orientation, and they only affect the non-cortical portions of the artifacts. Larger scars are found on artifacts made from the coarse-grained flint variety, where they occur in isolation along the edges (Fig. 16D, 16E, 16F), but are less numerous (spanning from three scars on specimen T1 up to 14 on specimen T11) (Table 4). In contrast, artifacts made from the fine-grained flint tend to present more scars, which are smaller in size and frequently clustered along the edges (Fig. 16G, 16H, 16L). Furthermore, the patina on the coarse-grained flint artifacts overlaps the mechanical microflaking scars, providing evidence for a relative

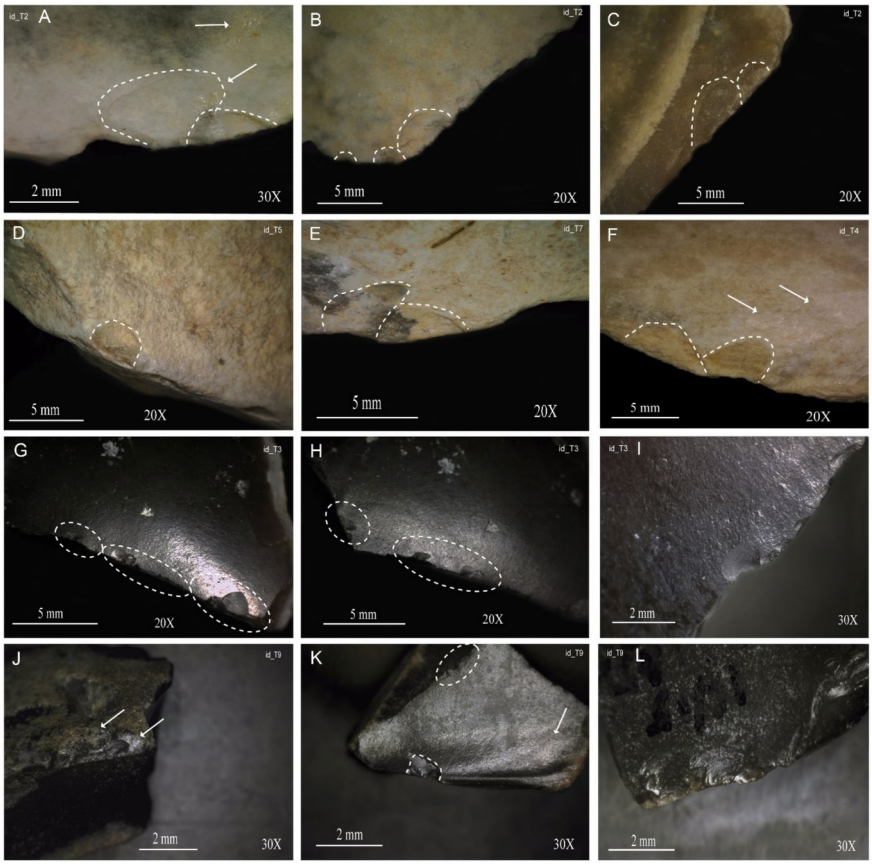

Fig. 16 Selection of photographs showing PDSM observed on the archaeological sample. Dotted lines indicate the microflaking scars; arrows indicate the presence of polish. Microflaking scars on coarse-grained flint are larger and mostly isolated (**A**, **B**, **C**, **D**, **E**, **F**), while on fine-grained flint, they are usually smaller and grouped (**G**, **H**, **I**, **L**) (see text for details). The figure also illustrates different degrees of patina and rounding, from low (**G**, **H**, **I**) to medium (**A**, **B**, **C**, **J**, **K**, **L**), and high (**D**, **E**, **F**)

chronology of the alteration processes affecting the archaeological material (*e.g.*, Berruti & Arzarello, 2020).

Rounding was observed on the whole assemblage. On the fine to very fine-grained flint artifacts, rounding is mild and affects mainly the edges and the prominent surfaces (barely affecting the ridges). Edge rounding is also observable on the coarse-grained flint artifacts, whereas ridge rounding was not fully assessable, due to the heavy patina covering them. Nevertheless, it is likely that the processes of patina formation of the coarse-grained flint artifacts also contributed to the heavy rounding of the edges and ridges, as also observed during our chemical experiment. Finally, polish was observed on localized portions of artifacts T2, T4, and T9 (Table 4, Fig. 16A, 16F, 16J).

Discussion

The methodological and sequential experimental framework presented in this study highlights how different flint types respond to mechanical and chemical stresses, demonstrating the potential of the approach for broader application. This study enabled us to formulate a preliminary and cautious hypothesis regarding the taphonomic processes at the Wadi Lazalim case-study site. The results thus provide a basis for future quantitative work on assemblage integrity at the site and may serve as a reference for further analysis. By simulating hydraulic transport, we identified six different PDSM that formed at different stages of the experiment (Fig. 6). Some alterations, such as the detachment of thin chips of material—difficult to detect on archaeological artifacts—tend to develop during the initial stages, alongside edge fractures. In contrast, microflaking, rounding, and polish increased progressively over time, as also documented by previous studies (Asryan *et al.*, 2014; Burroni *et al.*, 2002; Mazzucco *et al.*, 2013) (Fig. 6). Raw material differences, as well as the artifacts' edge thickness, also strongly influence the development of mechanical PDSM. As expected, thinner and more prominent surfaces exhibit more intensive alterations than thicker portions (Asryan *et al.*, 2014; Burroni *et al.*, 2002) (including ridges and cortical portions). This is particularly the case for abrasive polish and bright spots, which were mainly located on the more protruding areas, often near ridges, edges, or previous scars (Fig. 11). However, as time increases, even the thickest and most robust surfaces, such as cortex and ridges, undergo alteration. Our experiments also show that when sediment granulometry, mechanical action, and exposure time are held constant, artifact density plays a pivotal role in the development and intensity of PDSM. In fact, the higher the density of artifacts transported, the faster and the heavier the development of the alterations. This may have important implications when studying high-density assemblages. For instance, at the high-density artifact site of Arbo, NW Spain, the authors found that, despite substantial accumulation of artifacts along the sequence by water transport, altered materials were relatively few, and alteration degrees mainly depended on different raw materials and artifact size (Méndez-Quintas *et al.*, 2022). Although an in-depth PDSM analysis is not provided by the authors, they conclude that, despite clear evidence of disturbance, the technological characteristics of the assemblages were coherent,

thereby supporting the overall integrity of the assemblage. Similarly, the site of Garba I in Ethiopia is characterized by the accumulation of more than 12,000 lithic elements in a dynamic fluvial context (Galán *et al.*, 2022). The macroscopic analysis of 1678 artifacts revealed extensive surface alteration and edge damage, which is thought to be linked to the high density of artifact accumulation rather than to intense sedimentary disturbance (Galán *et al.*, 2022). Our results, if systematically applied to contexts characterized by high-density artifact accumulations, may offer valuable insights for more accurately assessing assemblage integrity, minimizing (or not) the degree of site disturbance.

Moreover, we did not find any correlations between the morphometric characteristics of the artifacts and the development of surface alterations. Although our experiment was not aimed at evaluating the effects of sediment granulometry on the development of PDSM, previous studies have reported a correlation between sediment size and degree of alteration, with coarser sediments resulting in more intensive alterations (Bustos-Pérez *et al.*, 2019; Chu *et al.*, 2015). We also documented the process of patina formation at different stages by measuring changes in surface roughness and correlated the degree of patina formation with variations in raw material texture. These results were subsequently compared with a sample of archaeological materials. PDSM were observed on the entire archaeological sample, and their characteristics match those recorded on the experimental specimens and with common tribological features (*e.g.*, Burroni *et al.*, 2002; Mate, 2008). Moreover, the relative proportions of recorded alterations confirm the experimental observation that fine-grained flint is more prone to edge scarring, whereas edge rounding affects coarse-grained artifacts to a much greater extent, probably indicating that chemical processes are also involved in edge rounding. This indicates that archaeological lithic materials from site 16/15 at Wadi Lazalim underwent some degree of remobilization. However, mechanical alterations are generally limited. PDSM are equally distributed on the brownish and darkish flint artifacts, regardless of their morphometrical characteristics. The moderate extent of the alterations may suggest either a short time of transport and/or a low-energy depositional environment for artifact remobilization (Petraglia & Potts, 1994). Moreover, artifacts recovered from layers G, H, and I show similar percentages of various PDSM (Fig. 15). These layers are characterized by consolidated sandy-silty deposits containing similar proportions of rare pebbles and gravels ranging from 1 to 30 cm. The juxtaposition of coarse sediments (gravel and lithics), interspersed with mainly fine sands, has been interpreted as the result of an alluvial/colluvial accumulation with sedimentation occurring at different stages (Cancellieri *et al.*, 2022), likely indicating a low-energy environment (Hosfield & Chambers, 2005; Petraglia & Potts, 1994). The overall homogeneity of PDSM observed on artifacts from these layers, both on fine- and coarse-grained flints, may further suggest similar deposition events. However, slightly larger scars on artifacts from layer I (Table 4), which also contains the highest concentration of artifacts (see Cancellieri *et al.*, 2022), may reflect the influence of coarser sedimentary components within its matrix, maybe suggesting a slightly higher-energy environment. In addition, the variability in the recorded PDSM may be closely related to the fine matrix of these layers (Bustos-Pérez *et al.*, 2019; Bustos-Pérez & Ollé, 2024a). By contrast, artifacts collected from layers B and K display lower damage.

Although the sample analyzed needs to be expanded, a greater contribution of pedogenetic processes has been suggested at least for the top of the sedimentary sequence (Cancellieri *et al.*, 2022).

The overall fresh condition of the finer-grained flint artifacts from site 16/15 may indicate limited transport, both in distance and duration, as experimental evidence shows that this material is particularly prone to rapid and extensive alteration under mechanical stress. Conversely, there are great differences in the chemical alteration of the archaeological artifacts. Coarse-grained artifacts exhibit a high degree of surface alteration, characterized by a widespread, well-developed white patina. Medium- to fine-grained flint artifacts were moderately affected by these processes (less diffuse and thinner white patina), whereas very fine-grained flint artifacts show moderate to no patina at all (see also Cancellieri *et al.*, 2022). Chemical experiments indicate that fine-grained flint is more susceptible to alteration in basic solutions but resists acidic environments, likely due to its strongly silicified structure. On the contrary, coarse-grained flints are more prone to alteration than the fine-grained varieties, even within less alkaline soils, likely because of their different mineral and fossil compositions that may involve a greater dissolution rate of the non-siliceous component. Observations from both archaeological and experimental artifacts indicate that the formation of white patina, prompted by an elevated pH in alkaline soils and enhanced by water (Caux *et al.*, 2018; Fiers, 2020; Glauberman & Thorson, 2012; Rottländer, 1975), was a major factor in artifact surface alteration at the Wadi Lazalim site, predominantly affecting the coarse-grained artifacts. Based on the analyzed sample of archaeological materials, we hypothesize that the lithic material from site 16/15 may have been deposited by gentle sediment movement caused by runoff processes activated by water, which could have transported over short distances a high density of artifacts accumulated on the top of the deflated stone pavements within a generally gently sloping landscape. This may be inferred from the tribological stigmata observed on the artifact's surfaces, such as the development and characteristics of microflaking, rounding, and polish, the latter resulting from a combination of both mechanical and chemical processes, enhanced by water (Berruti & Arzarello, 2020; Burroni *et al.*, 2002; Levi-Sala, 1986; Wiederhold & Pevny, 2014). Moreover, extensive mechanical alterations, common on lithics transported in high-energy fluvial systems (Grosman *et al.*, 2010; Petraglia & Potts, 1994; Shea, 1999), are absent from our assemblage. The overall homogeneity of mechanical PDSM observed on artifacts belonging to different archaeological layers (same degree of mechanical alteration), as well as the absence of artifact size sorting (Cancellieri *et al.*, 2022)—a common situation in high energy contexts/long distance fluvial transport (Bertran *et al.*, 2012; Byers *et al.*, 2015; Chu & Hosfield, 2020; de la Torre *et al.*, 2018; Petraglia & Potts, 1994)—may be indicative of assemblage integrity. Furthermore, differences in surface alteration, as highlighted by our experiment, are predominantly driven by chemical processes that strongly depend on raw material characteristics, such as texture and porosity. While coarser flints resist better to mechanical stresses, they are more vulnerable to chemical alteration than the fine-grained ones (Burroni *et al.*, 2002; Fiers, 2020; the only exception is a change in color due to UV exposure, which appears to affect only fine-grained flints, Michel & Rots, 2022). In our assemblage, chemical rather than mechanical

processes appear to be the primary factor of the co-occurrence of differently altered materials. This evidence, coupled with the fact that patina covers the mechanical microflaking scars, may indicate that the artifacts were first shortly transported, and that, subsequently, chemical alteration started to develop before their burial and/or after it, due to diagenetic processes and strongly depending on the depositional context rather than erosional activities. The artifacts were likely buried in alkaline, relatively moist soils, where the presence of alkaline materials and occasional wetting would have promoted patina formation (Howards, 2002). Their accumulation probably resulted from cyclic surface runoff, erosional, and depositional processes during semi-arid conditions throughout the interglacial/glacial alternating phases of the late Middle Pleistocene (Zerboni *et al.*, 2011). Although the analysis of a larger sample of archaeological artifacts is required to confirm or refute this hypothesis, geomorphological and stratigraphic data, coupled with the experimental results, constitute a valuable basis for further investigating the taphonomic processes at the site.

More experimentation is also needed. For instance, it remains unclear whether patina development may hinder the mechanical alteration, as it seems to be partially the case in the analyzed archaeological sample, a pattern also highlighted in other contexts (Galán *et al.*, 2022). This is particularly relevant for edge/ridge rounding. Burroni *et al.* (2002) reported that artifacts tumbled in wet sediments exhibit a faster and more pronounced degree of ridge rounding than those tumbled in dry sediments. Such effects are likely the result of a chemical process, promoting the development of a hydrated film on the surface that enhances micro-cracking, as previously observed in ceramics (Kato, 1990). Moreover, the dissolution and reprecipitation of siliceous constituents on artifact surfaces, leading to patina formation, may reduce surface microtopography and increase smoothness through abrasion-accelerated dissolution (Howards, 1999, 2002; Thiry *et al.*, 2014). Our experiments also show that artifacts transported in wet sediments develop edge/ridge rounding from the earliest stages, followed by a steady phase and then by a rapid and increased rounding over time. A similar pattern is observed in the development of polish and bright spots, regardless of artifact density, potentially indicating a contribution from chemical processes enhanced by water (Grace, 1996). Likewise, during the chemical experiment, we observed the development of both edge and ridge rounding, as well as edge fracture and flaking, demonstrating that chemical and mechanical alterations are intimately interrelated phenomena. A further step in the analysis should test how these processes interact and to what extent the development of one can hinder the recognition of the other (by tumbling previously chemically altered artifacts and vice versa, for instance). Moreover, recent research has shown that the development of sedimentary abrasion, which can result in the formation of polish and bright spots, is related to artifacts' topography and that it usually develops faster and more intensely on convex surfaces (Bustos-Pérez & Ollé, 2024a). By using gray-level images of the surfaces to extract quantitative data on surface alteration, the authors were able to differentiate between times of exposure and to show that the heterogeneous development of sedimentary abrasion is linked to both flake topography and sediment particle size (Bustos-Pérez & Ollé, 2024a). Although we did not apply quantitative approaches to assess degrees of alteration, we documented the development of abrasive polish and bright spots, primarily on the more prominent

surfaces of the artifacts (except for the bulbs). These were most frequently observed near ridges, edges, or previous scars, in both the experimental and archaeological samples (Figs. 11 and 16). The absence of alteration on the bulbs may be related to the lower rotation speed of the tumbling machine used in our experiment compared to that employed by Bustos-Pérez and Ollé (2024a), since both experiments shared similar sediment granulometry and exposure time. Further experimentation, coupled with a quantification of the resulting PDSM (*e.g.*, Bird *et al.*, 2007; Bustos-Pérez *et al.*, 2019; Bustos-Pérez & Ollé, 2024a, 2024b; Grosman *et al.*, 2010; Hosfield *et al.*, 2000; Schoville, 2010; Shackley, 1974), will enhance and refine these observations.

Finally, the use of different microscopes for the analysis of the archaeological and experimental samples, although probably not affecting the overall observation patterns, may have hindered the identification of some specific alterations, notably at higher magnifications.

Conclusion

Although time-consuming, understanding how lithic artifacts respond to taphonomic processes has proven highly valuable for inferring site formation processes and assemblage integrity. Moreover, many Palaeolithic/Stone Age sites' assemblages were accumulated or disturbed by water flow, highlighting the importance of such analyses as an essential step in interpreting site formation processes. To evaluate the extent to which different flint types, as recognized in the archaeological record of site 16/15 at Wadi Lazalim, react to both mechanical and chemical stresses, we established a methodological and experimental framework. Alterations observed in the experimental sample indicate a differential taphonomic behavior between fine-grained and coarse-grained flints, while also demonstrating that multiple variables influence the degree of PDSM development. These results provide a solid foundation for approaching the analysis of the archaeological assemblage of site 16/15. Preliminary comparisons indicate that chemical processes likely played a greater role than mechanical ones in shaping the archaeological assemblage and that differences observed in the surface alteration degrees are likely to be linked to differences in the raw material texture. These experiments highlight the variability in resistance to alteration between different flint types and offer a cautious basis for hypothesis-building and future investigations. Expanding the sample size and applying quantitative analyses to the study of PDSM will significantly strengthen these observations and allow for a more refined interpretation of the taphonomic and formation processes.

Acknowledgements The research is part of the activities of The Archaeological Mission in the Sahara, Sapienza University of Rome, directed by Savino di Lernia. The international research project in Tunisia was directed by N. A. (INP), Lotfi Belhouchet (INP), and Savino di Lernia. We sincerely thank the Italian Embassy and the Italian Institute of Culture in Tunis for their support and assistance. Special thanks to the Director of the Institut National du Patrimoine and the institution's staff for their indispensable support during our work in Tunisia. We also thank Amir Gharbi, Martina Di Matteo and Caterina Aureli for their help during the geological sampling and the recording of the experimental knapping sequences. Finally, we would also like to thank the anonymous reviewers for their thorough and useful comments.

Author contribution O.S. wrote the paper with contributions from all authors. O. S., C. L., and A. Z. conceived and set up the mechanical experiment. O. S. and C. L. carried out the mechanical experiment and analyzed the results. O. S. and A. Q. carried out the chemical experiment and analyzed the results. O. S. and E. C. analyzed the archaeological materials. H. M. provided support for the excavations and for the archaeological analyses. T. B. F. provided geomorphological and sedimentological data. E. C. directed the excavations at the site and realized the experimental artifacts for this research. Savino di Lernia and N. A. designed the research, directed the fieldwork, and granted access to archaeological data. The authors equally contributed to the discussion.

Funding Open access funding provided by Università degli Studi di Roma La Sapienza within the CRUI-CARE Agreement. Funds from Sapienza University of Rome (Grandi Scavi di Ateneo), Italian Ministry of Foreign Affairs and International Cooperation (DGSP), and the National Geographic Society-Waytt Institute were entrusted to Savino di Lernia. Further resources from the Wenner Gren Foundation and the Leakey Foundation were entrusted to E.C. E.C. is currently supported by an ERA postdoctoral fellowship funded under the Horizon Europe – Widening participation and spreading excellence program (Grant agreement ID 101180691).

Data availability All results not provided here, as well as the underlying data, programming code, and the figures of the alteration presented in the text, are available in an online repository athttps://doi.org/10.5281/zenodo.17076226.

Declarations

Competing interests The authors declare no competing interests.

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